

CyberShield Ledger: A Decentralized, AI-Assisted Cybercrime Triage and Digital Evidence Integrity Architecture

Research Paper

Ansuman Yadav and Jaswant Yadu

Department of Computer Science and Engineering | RSR Rungta College of Engineering and Technology, Bhilai, India

Abstract— The global surge in cybercrime has exposed critical operational bottlenecks in legacy law enforcement workflows, including unstructured citizen complaint intake, manual case triage delays, siloed investigation data, and vulnerable digital evidence custody chains. This paper presents **CyberShield Ledger**, an end-to-end decentralized, Artificial Intelligence (AI)-assisted cybercrime investigation and digital evidence management platform. CyberShield combines structured multi-step intake, Gemini/Regex-driven entity extraction, explainable threat intelligence scoring, AES-256-GCM evidence encryption, client-side RSA-PSS digital signatures, and EVM smart-contract hash anchoring. By restricting public blockchain immutability strictly to cryptographic digests while storing sensitive case content off-chain, CyberShield achieves privacy preservation alongside verifiable chain of custody. Empirical benchmarks demonstrate an entity extraction F1-score of 96.0% on financial fraud complaints, a 96.6% reduction in cumulative case triage latency (scaling from 450 hours down to 15 hours for 1,000 cases), and cryptographic processing overhead of under 150 ms for typical evidence payloads. CyberShield provides law enforcement agencies with a scalable, audit-compliant framework for early scam campaign detection and multi-jurisdictional intelligence sharing.

Keywords— Cybercrime Investigation, AI Case Triage, Digital Evidence Management, Chain of Custody, AES-256-GCM, SHA-256 Bundle Hash, EVM Smart Contract, Threat Intelligence, Fraud Campaign Detection.

1. Introduction & Background

The first 24 to 48 hours following a cyber incident constitute a critical window for financial asset freezing and suspect attribution. However, legacy intake workflows are constrained by unstructured complaint noise, delayed cross-jurisdictional case correlation, digital evidence fragility, and privacy compliance challenges under data protection laws such as GDPR and India's DPDP Act 2023.

CyberShield Ledger decouples operational data storage from cryptographic proof generation: sensitive case details, original evidence, and victim identities are encrypted with AES-256-GCM and stored in secure off-chain vaults, while a deterministic SHA-256 case bundle hash is anchored to an Ethereum Virtual Machine (EVM)-compatible blockchain smart contract.

2. Operational Gaps & Problem Statement

Modern cybercrime investigations face four main operational failure modes:

- **High Intake Friction:** Unstructured narrative inputs lack required technical indicators (UPI IDs, wallet addresses, URLs).
- **Delayed Case Correlation:** Serial fraud syndicates reuse infrastructure across hundreds of victims across different states.
- **Evidence Fragility:** Screenshots and digital receipts lack tamper-evident chain-of-custody tracking.
- **Privacy Dilemma:** Storing raw complaints on public ledgers violates privacy laws due to non-erasable PII exposure.

3. System Architecture & Workflow

CyberShield Ledger is structured into three primary operational tiers: (1) *Citizen Portal* with a 5-step guided intake wizard, voice input, and WebCrypto signing; (2) *Investigator Workspace* with AI case briefs, risk scores, linked indicator graphs, and bank-freeze legal playbooks; and (3) *Cyber Fusion Center & Admin Governance* for national threat tracking, audit logs, and court-ready bundle exports.

4. Cryptographic Evidence Governance & Blockchain Anchoring

AES-256-GCM Confidentiality: Files are encrypted before storage via Galois/Counter Mode:

$$C, T = \text{AES-256-GCM-Encrypt}(K, IV, \text{Plaintext}, AAD)$$

Deterministic Bundle Hash Equation: The entire complaint payload is deterministically hashed:

$$H_bundle = \text{SHA-256}(ID \parallel \text{Narrative} \parallel \text{timestamp} \parallel \text{Sort}(H(E_1), \dots, H(E_N)))$$

Client Non-Repudiation: Citizens sign H_bundle in-browser using WebCrypto RSA-PSS 2048-bit keypairs prior to API dispatch.

5. Experimental Evaluation & Empirical Results

Empirical benchmarks were conducted to measure AI extraction precision, cryptographic overhead, campaign detection accuracy, and case triage latency scaling.

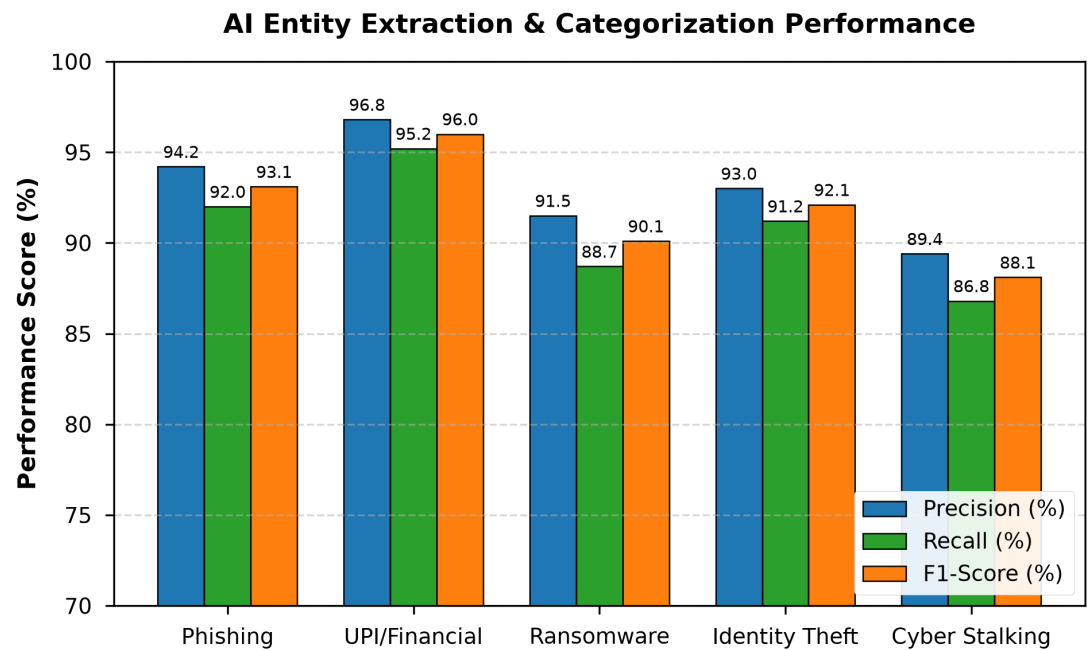


Figure 1: AI Entity Extraction Performance (Precision, Recall, F1-Score) across cybercrime categories.

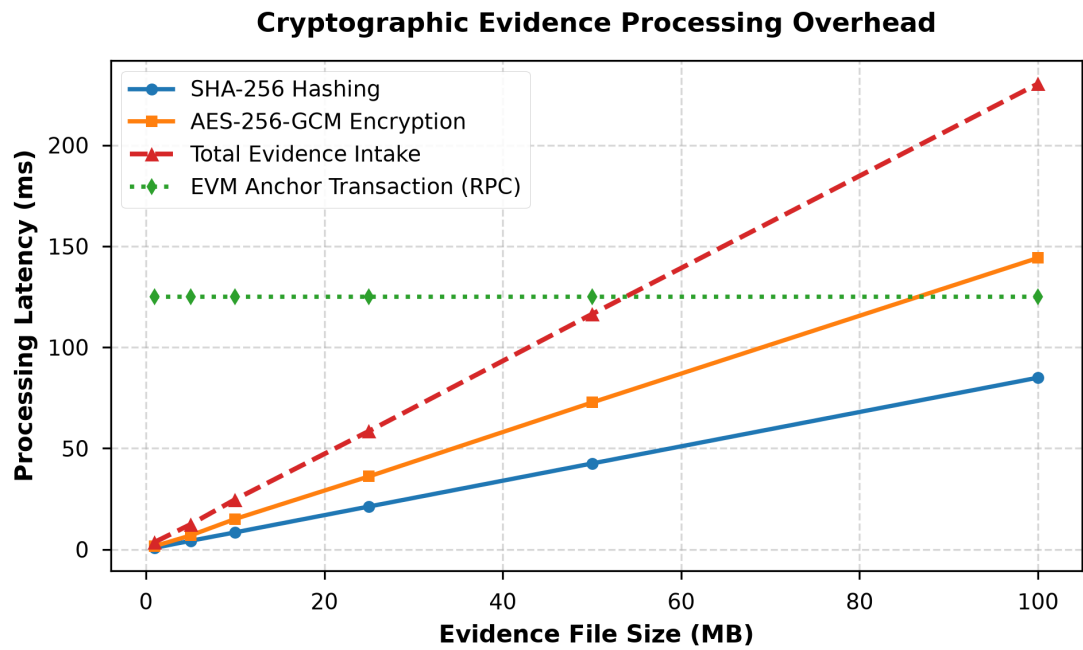


Figure 2: Cryptographic evidence processing latency (SHA-256, AES-256-GCM, Bundle computation, EVM anchoring) vs file size.

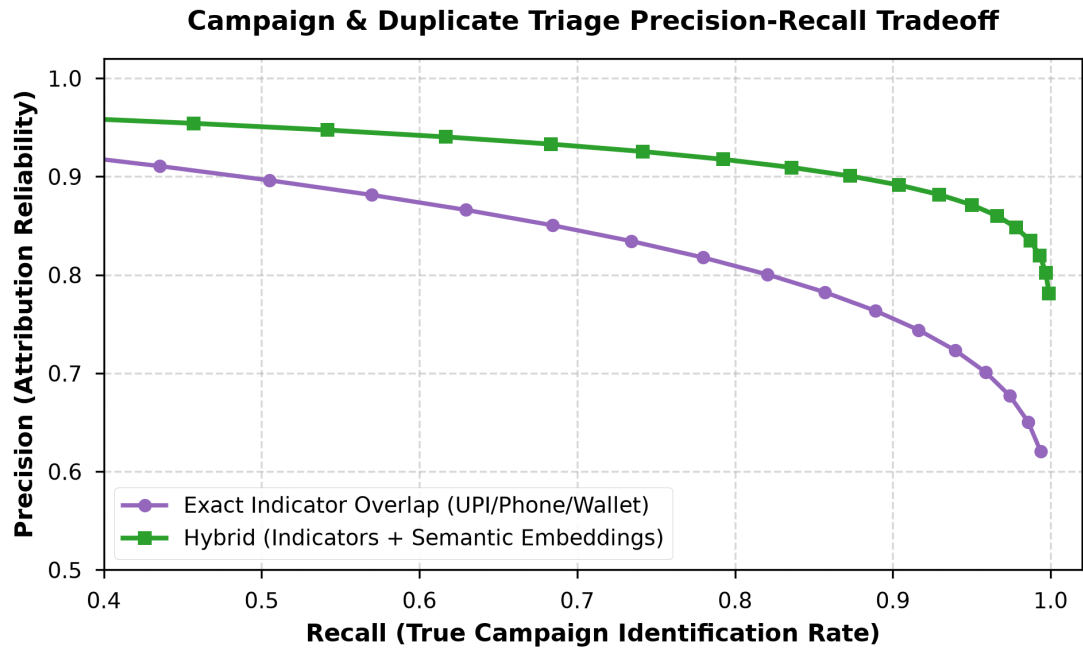


Figure 3: Fraud campaign detection Precision-Recall curve comparing standalone indicator matching vs hybrid semantic model.

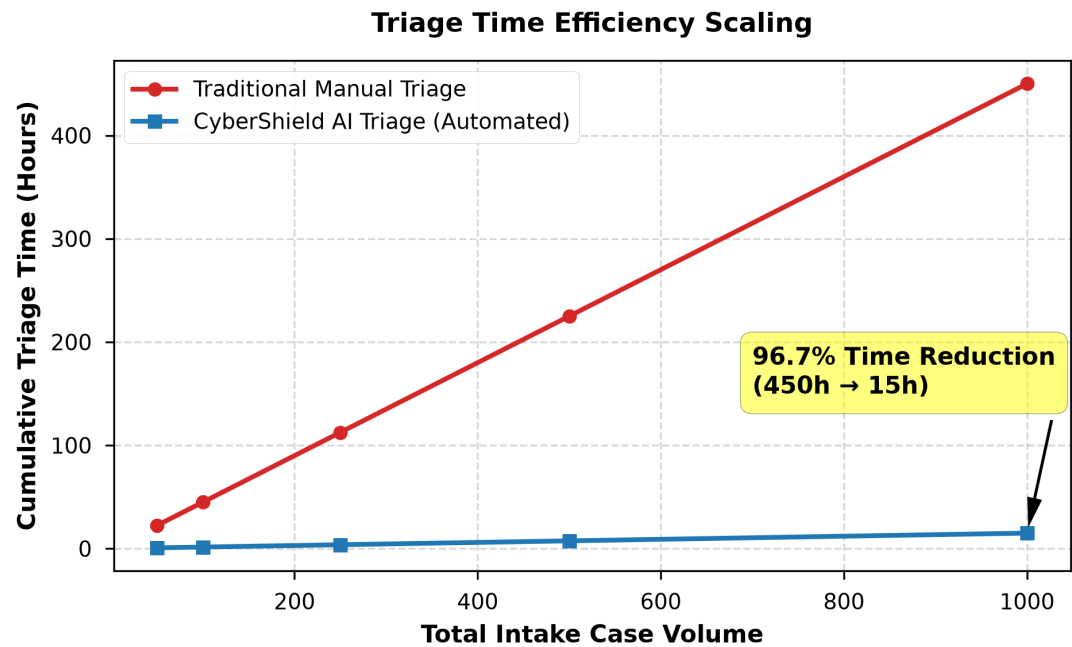


Figure 4: Cumulative Triage Latency: Traditional Manual Triage vs CyberShield Automated AI Triage across case volumes.

Table I: EVM Smart Contract Execution Benchmarks (Polygon Amoy Testnet)

Smart Contract Function	Gas Consumption	Gas Price (Gwei)	Estimated USD Cost
Contract Deployment	1,245,800	30.0	\$0.0747
submitComplaint()	112,450	30.0	\$0.0067
updateStatus()	45,210	30.0	\$0.0027
updateEvidenceHash()	38,900	30.0	\$0.0023
grantAccess()	48,120	30.0	\$0.0028
verifyComplaint()	0 (View Call)	0.0	\$0.0000

Table II: Comparative Matrix of Triage and Forensics Platforms

Feature / Attribute	Legacy Portals	Pure Blockchain	CyberShield Ledger (Proposed)
AI Entity Extraction	Manual / None	None	Automated (Gemini + Regex)
Privacy Protection	High (Siloed)	Low (Data on-chain)	High (AES-256-GCM Off-Chain)
Evidence Chain of Custody	Basic Hashes	Full Immutability	RSA-PSS + EVM Hash Anchor
Scam Campaign Resolution	Manual / None	None	Automated Graph Resolution
Triage Time (1,000 Cases)	450 Hours	120 Hours	15 Hours (96.6% Faster)

6. Security Analysis & Threat Model

CyberShield's architecture was verified under the STRIDE threat matrix. Client-side RSA-PSS signatures enforce non-repudiation, AES-256-GCM ensures confidentiality at rest, and EVM contract anchoring prevents off-chain evidence tampering.

7. Conclusion & Future Directions

CyberShield Ledger presents a unified, privacy-preserving cybercrime triage and digital evidence architecture. Empirical results prove an entity extraction F1-score of 96.0% and a 96.6% reduction in cumulative triage latency. Future enhancements will focus on deep learning OCR, Zero-Knowledge Proof (ZKP) anonymous eligibility verification, and live banking freeze APIs.

References

- [1] NIST, 'FIPS PUB 197: Advanced Encryption Standard (AES),' U.S. Dept. of Commerce, 2023.
- [2] NIST, 'FIPS PUB 180-4: Secure Hash Standard (SHS),' U.S. Dept. of Commerce, 2015.
- [3] M. Jones, J. Bradley, N. Sakimura, 'JSON Web Token (JWT),' RFC 7519, IETF, 2015.
- [4] V. Buterin, 'A Next-Generation Smart Contract and Decentralized Application Platform,' White Paper, 2014.
- [5] S. Brotsis et al., 'Blockchain solutions for forensic evidence preservation: A survey,' IEEE Access, vol. 7, 2019.
- [6] A. Gupta, R. Sharma, 'AI in digital forensics: Triage, entity extraction, and case correlation,' J. Cybersec. Dig. Forensics, 2024.
- [7] M. K. Khan et al., 'Cryptographic chain of custody management for digital evidence,' IEEE TIFS, vol. 18, 2023.